

Transportation Data Analysis and Traffic Simulation

NHTS Travel Behavior, NGSIM Vehicle Trajectories, and
Intelligent Driver Model Simulation

Project #3 | CIVE 202 – Civil Engineering Analysis II
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Prepared for: Federal Highway Administration (FHWA) client prompt
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Executive Summary

This project analyzes transportation behavior at two different scales using real-world transportation datasets supplied for a Federal Highway Administration (FHWA) client prompt. The National Household Travel Survey (NHTS) dataset was used to examine vehicle characteristics and household context, while the Next Generation Simulation (NGSIM) dataset was used to study vehicle trajectories and car-following behavior. The project then implemented the Intelligent Driver Model (IDM) in Python and compared a simulated follower trajectory with observed NGSIM data.

The final notebook demonstrates a complete Python workflow: loading external CSV data, selecting and filtering variables, creating categorical and distribution visualizations, analyzing time-series vehicle trajectories, defining a reusable simulation function, and using Euler's method to update vehicle speed and position over time. The analysis therefore connects descriptive transportation data analysis with microscopic traffic-flow modeling.

The NHTS dataset contained 14,684 vehicle-level records. Passenger automobiles represented 43.46% of the records, SUVs 31.07%, and pickup trucks 16.30%; together these three categories represented approximately 90.83% of the sample. Vehicle-age analysis showed an average age of 11.72 years for rural records compared with 9.67 years for urban records, although the unequal sample sizes (3,600 rural versus 11,084 urban) mean raw histogram counts should not be interpreted as equal-population frequencies.

For the NGSIM component, the final notebook selected trajectory pair 1, containing 841 observations at 0.1-second intervals from 0.1 to 84.1 seconds. The observed leader-follower gap averaged 23.60 m and ranged from 10.36 m to 32.53 m. The leader and follower had similar mean speeds of 7.44 m/s and 7.38 m/s, respectively, which is consistent with a follower responding to the same traffic stream while maintaining separation.

The IDM simulation used a desired speed of 30 m/s, minimum spacing of 2 m, desired time headway of 1.5 s, maximum acceleration of 1.0 m/s^2 , comfortable deceleration of 1.5 m/s^2 , and acceleration exponent of 4. Using the same leader trajectory as input, the simulated follower captured the overall movement pattern of the observed follower. For report-level evaluation, the simulated position had an RMSE of approximately 9.91 m and a mean absolute error of 8.66 m; the final simulated position was about 5.30 m ahead of the observed follower. These results show that the IDM reproduced the broad trend but not every short-term fluctuation in human driving behavior.

From a transportation engineering perspective, the project demonstrates how household travel data, trajectory data, visualization, and microscopic simulation can support planning, traffic-flow analysis, model calibration, and future connected or automated vehicle studies. It also demonstrates that modeling decisions—such as selecting one trajectory, choosing IDM parameters, using raw histogram counts, or applying numerical safeguards—can influence the results and must be documented transparently.

1. Project Context and Objectives

The project goal was to analyze and visualize transportation data using NHTS and NGSIM datasets for the FHWA client prompt, develop data-driven insights into travel behavior and driving dynamics, and implement the Intelligent Driver Model to compare simulated car-following behavior with real-world traffic data.

The project addressed two complementary transportation questions. At the broader travel-behavior scale, the NHTS analysis asked what kinds of vehicles appear in the sample, how vehicle age differs between urban and rural households, and how vehicle age varies by manufacturer. At the microscopic traffic scale, the NGSIM analysis asked how a leader and follower move over time, how their spacing changes, and whether a mathematical car-following model can reproduce the observed follower trajectory.

- Use real transportation data from NHTS and NGSIM rather than fabricated examples.
- Create visualizations appropriate for categorical, distributional, and time-series data.
- Define and use a Python function for the Intelligent Driver Model acceleration equation.
- Use Euler's numerical method to simulate follower speed and position at 0.1-second time steps.
- Compare the simulated trajectory with observed NGSIM data and interpret the model's strengths and limitations.
- Connect the results to transportation planning, traffic operations, and future engineering analysis.

2. Data Sources and Variables

2.1 National Household Travel Survey (NHTS)

The NHTS file used in this project contained 14,684 records and 23 columns. The broader scope identified variables such as household income, vehicles per household, travel day, household location, and fuel type as potentially useful for travel-behavior analysis. The final notebook's implemented visualizations focused primarily on vehicle type, vehicle age, household location, and vehicle make.

Variable	Role in Final Notebook	Engineering Interpretation
vehicle_type	Categorical count plot	Composition of vehicle types represented in the sample
vehicle_age	Histogram and boxplot	Distribution and variability of vehicle age
household_location	Urban/rural filter	Context for comparing vehicle-age patterns
make	Boxplot grouping	Variation in vehicle age by manufacturer

The variables used directly in the final figures were largely complete. Vehicle type, vehicle age, and household location had no missing values in the supplied file. Vehicle make had 116 missing values, so the make-based boxplot necessarily reflects the records for which make was available.

This matters because missing manufacturer information can slightly change the apparent distribution for some categories.

2.2 Next Generation Simulation (NGSIM)

The NGSIM file contained 8,166 observations, eight variables, and 16 trajectory pairs. The final notebook selected trajectory number 1 for detailed car-following analysis and IDM simulation. NGSIM variables included time, leader and follower positions, leader and follower speeds, leader and follower accelerations, and trajectory number.

Variable	Units	Use
Time	s	Simulation and plotting time axis
leader_position(m)	m	Observed leader trajectory
follower_position(m)	m	Observed follower trajectory and simulation benchmark
leader_speed(m/s)	m/s	Input to relative-speed and IDM calculations
follower_speed(m/s)	m/s	Observed follower state and initial simulation speed
leader_acc(m/s ²)	m/s ²	Available observed leader dynamics
follower_acc(m/s ²)	m/s ²	Available observed follower dynamics
trajectory_number	—	Selects an individual leader-follower pair

3. Methodology

3.1 Python Environment and Packages

The final notebook used pandas for CSV loading and tabular filtering, NumPy for array-based numerical calculations, Matplotlib for plotting, and Seaborn for statistical visualization and line plots. This combination allowed the analysis to move from raw tabular data to visual interpretation and then to numerical traffic simulation within a single workflow.

3.2 NHTS Visualization Workflow

The NHTS analysis used three complementary plot types. A count plot summarized the categorical distribution of vehicle type. An overlaid histogram compared vehicle-age distributions between urban and rural records. A boxplot compared vehicle-age distributions across vehicle makes. These plot types were selected because each answers a different question: category prevalence, distribution shape, and between-group variability.

3.3 NGSIM Time-Series Workflow

Trajectory number 1 was filtered from the NGSIM file. Two time-series visualizations were then created. The first plotted leader and follower positions against time to show how both vehicles progressed through the roadway segment. The second calculated the instantaneous spacing as leader position minus follower position and plotted the resulting gap over time. These visualizations provide an intuitive representation of car-following behavior before introducing the simulation model.

3.4 Intelligent Driver Model Formulation

The notebook defined a reusable IDM acceleration function. The desired dynamic spacing was calculated as:

$$s^* = s_0 + vT + (v\Delta v) / (2\sqrt{ab})$$

where v is follower speed, Δv is follower speed minus leader speed, s_0 is minimum spacing, T is desired time headway, a is maximum acceleration, and b is comfortable deceleration. The desired spacing was constrained so that it could not fall below s_0 .

Follower acceleration was then calculated using:

$$a_{\text{IDM}} = a [1 - (v/v_0)^\delta - (s^*/s)^2]$$

The implemented parameters were $v_0 = 30$ m/s, $s_0 = 2$ m, $T = 1.5$ s, $a = 1.0$ m/s², $b = 1.5$ m/s², and $\delta = 4$. These are modeling choices in the notebook rather than calibrated values estimated from trajectory 1.

3.5 Euler Time Integration

The NGSIM observations were spaced at 0.1-second intervals, so the simulation used $\Delta t = 0.1$ s. At each time step, the program calculated the current leader-follower gap and relative speed, evaluated the IDM acceleration, and updated follower speed and position using a first-order Euler method:

$$v_{i+1} = \max(v_i + a_i\Delta t, 0) \quad \text{and} \quad x_{i+1} = x_i + v_i\Delta t$$

Two numerical safeguards were used: the gap was constrained to at least 0.1 m, and simulated speed was constrained to remain nonnegative. These safeguards prevent division by zero, negative spacing, and physically impossible negative speeds, but they also represent explicit modeling decisions that influence the simulation.

4. NHTS Results and Interpretation

4.1 Vehicle-Type Composition

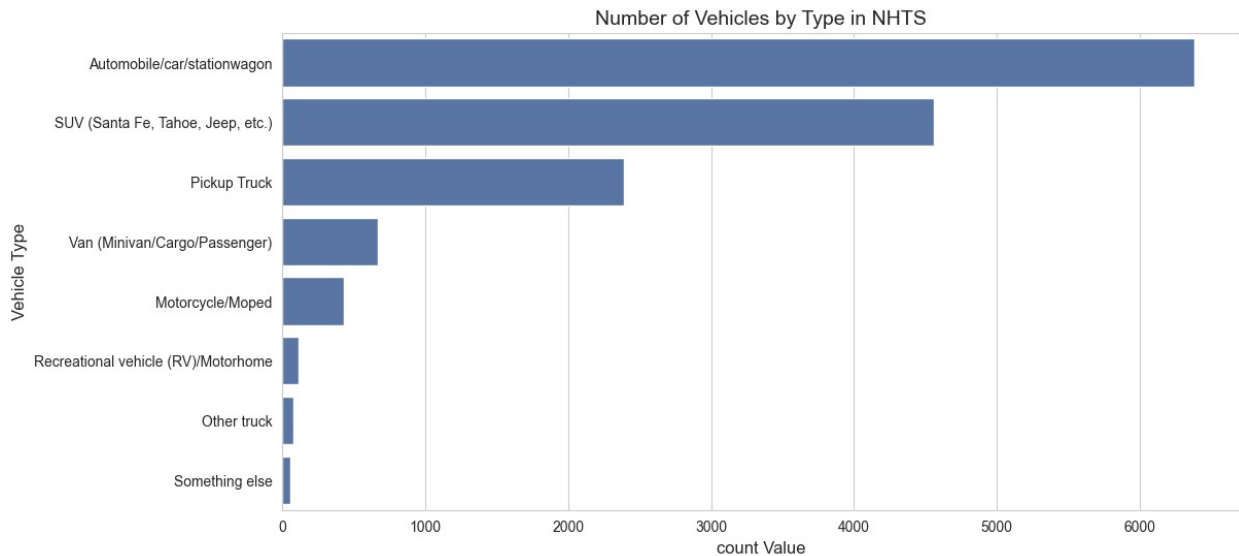


Figure 1. Number of Vehicles by Type in the NHTS dataset.

Passenger automobiles were the largest category with 6,382 records (43.46%), followed by SUVs with 4,562 records (31.07%) and pickup trucks with 2,393 records (16.30%). Together, these three categories accounted for approximately 90.83% of all records. Vans represented 4.57%, motorcycles/mopeds 2.93%, recreational vehicles 0.76%, other trucks 0.55%, and “something else” 0.36%.

The figure indicates that the sample is dominated by conventional personal vehicles rather than specialized vehicle categories. For transportation planning, the result is useful because vehicle mix affects roadway demand, parking needs, fuel consumption, emissions, and the types of facilities that households rely on. The plot should be interpreted as a description of the supplied NHTS sample rather than a direct national market-share estimate unless survey weights are explicitly incorporated.

Vehicle Type	Count	Share of Sample
Automobile/car/stationwagon	6,382	43.46%
SUV	4,562	31.07%
Pickup Truck	2,393	16.30%
Van	671	4.57%
Motorcycle/Moped	430	2.93%
RV/Motorhome	112	0.76%
Other truck	81	0.55%
Something else	53	0.36%

4.2 Urban and Rural Vehicle-Age Distribution

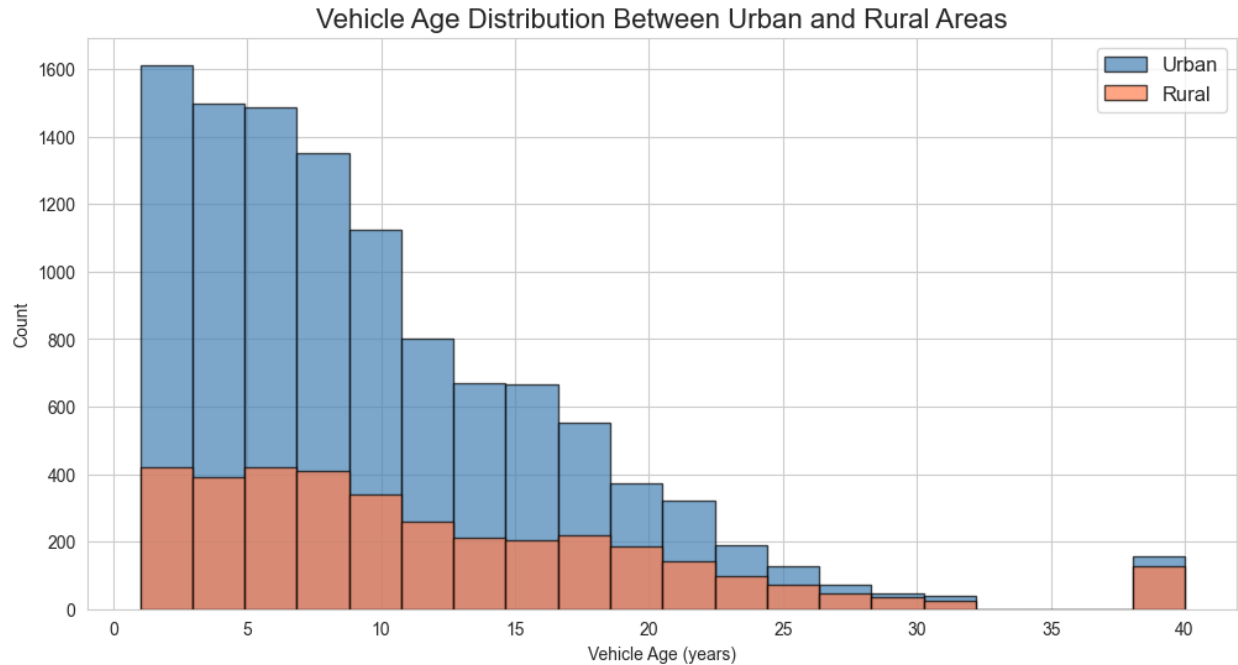


Figure 2. Vehicle-age distribution for urban and rural records.

The rural records had an average vehicle age of 11.72 years and a median of 9 years, while urban records had an average age of 9.67 years and a median of 8 years. The difference in mean age was approximately 2.05 years. Both groups included vehicles from 1 to 40 years old, but rural records showed greater variability, with a standard deviation of 9.05 years compared with 7.50 years for urban records.

This suggests that older vehicles are more prominent in the rural subset of the supplied dataset. A transportation engineer might consider several possible explanations—longer vehicle retention, different replacement cycles, lower transit availability, or different household travel needs—but the dataset alone does not establish causation. The analysis therefore supports a descriptive conclusion, not a causal one.

The plot also demonstrates an important visualization limitation: the histogram uses raw counts, but the sample contains 11,084 urban records and only 3,600 rural records. The taller urban bars partly reflect sample size. For a direct probability comparison between urban and rural distributions, normalized histograms or density plots would be preferable.

Household Location	Records	Mean Age (yr)	Median (yr)	Std. Dev. (yr)
Urban	11,084	9.67	8.0	7.50
Rural	3,600	11.72	9.0	9.05

4.3 Vehicle Age by Manufacturer

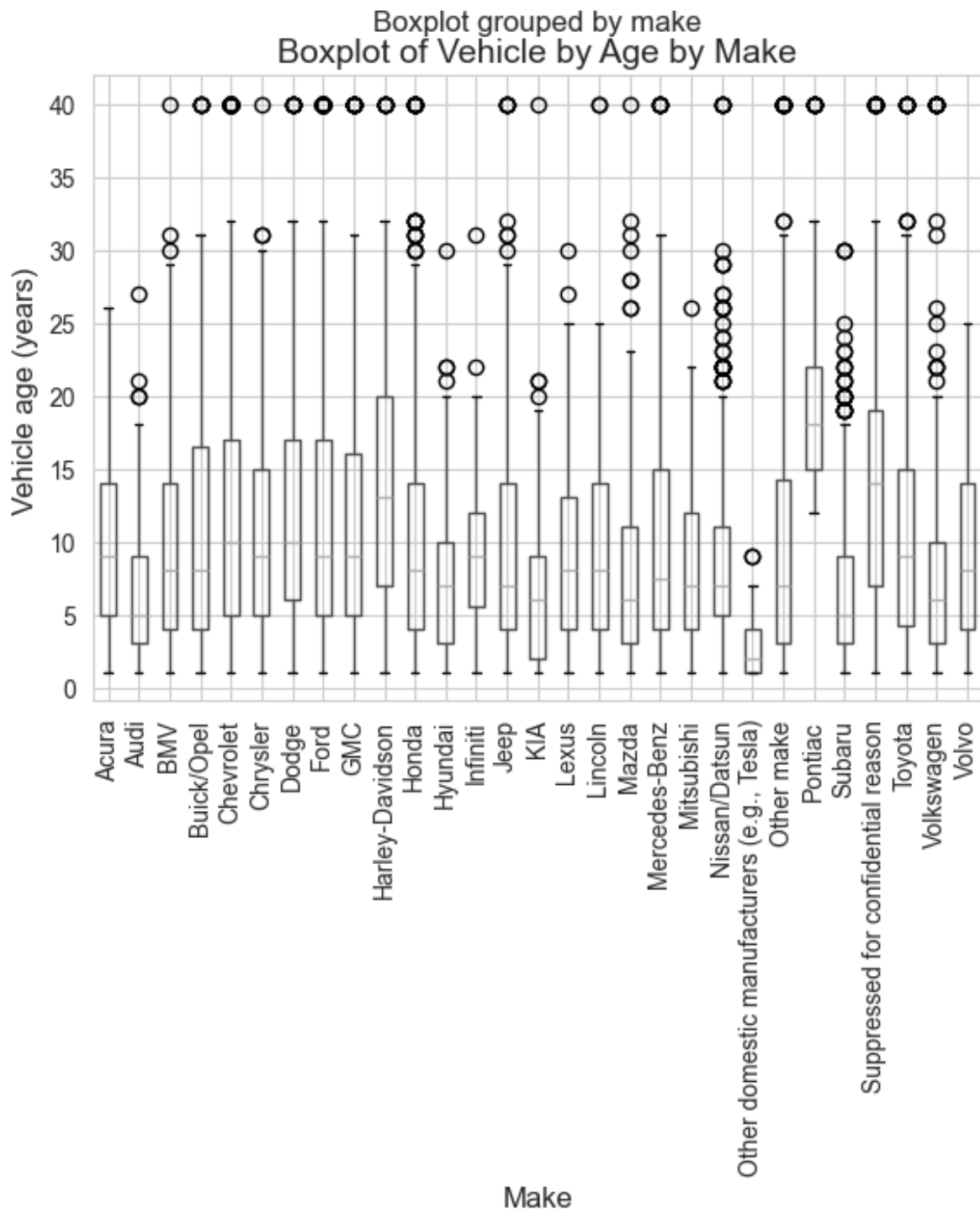


Figure 3. Boxplot of vehicle age by manufacturer.

The manufacturer boxplot shows that vehicle-age distributions vary considerably across makes and that most makes include a wide range of vehicle ages. Among the most represented makes in the dataset, Toyota had 1,974 records, Ford 1,921, Chevrolet 1,571, and Honda 1,507. Their mean ages were approximately 9.96, 11.62, 12.13, and 9.66 years, respectively. Subaru, Hyundai, and

KIA had younger average ages in this sample, while some long-established brands showed larger upper-age tails.

The boxplot is especially useful because it displays median, spread, and outliers simultaneously. However, the number of observations differs substantially by manufacturer, and 116 records were missing make information. Therefore, differences in box width are not shown, and the plot should not be interpreted as evidence that one manufacturer is inherently more durable than another. It describes the observed age distributions in the supplied NHTS file.

5. NGSIM Trajectory Results

5.1 Leader and Follower Position

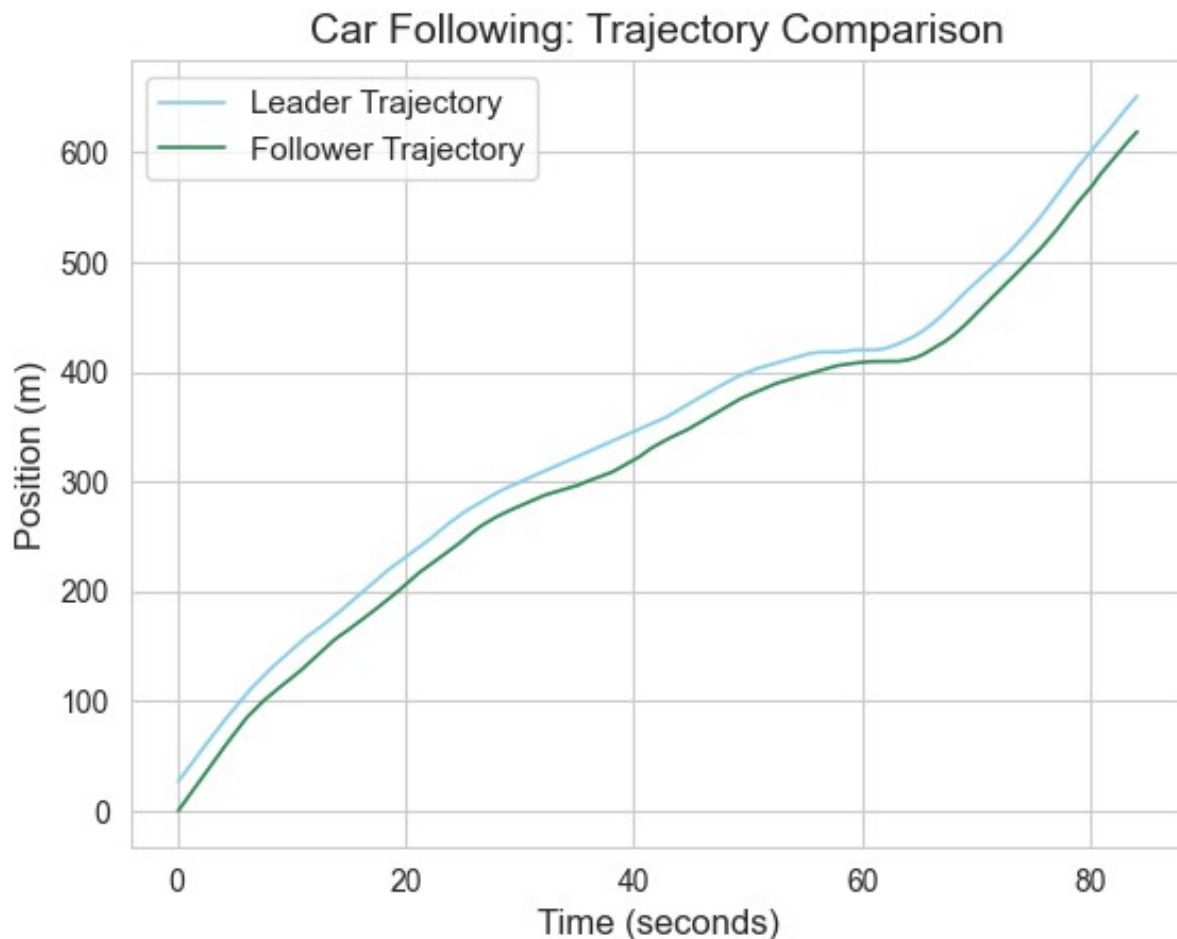


Figure 4. Observed leader and follower trajectories for NGSIM trajectory pair 1.

Trajectory pair 1 contained 841 observations from 0.1 s to 84.1 s. The leader and follower position curves move in parallel over much of the observation period, which indicates that the follower generally responds to the same traffic conditions while remaining behind the leader. The observed mean leader speed was 7.44 m/s and the observed mean follower speed was 7.38 m/s. The similarity of these mean speeds is consistent with a stable car-following relationship over the full observation period, even though short-term accelerations and decelerations differ.

The leader speed ranged from 0 to 15.18 m/s, and follower speed ranged from 0 to 16.26 m/s. This range indicates that the pair experienced substantial changes in traffic state rather than moving at a constant free-flow speed. Periods of low or zero speed are important for car-following modeling because spacing and relative speed become dominant drivers of acceleration decisions.

5.2 Gap-Distance Behavior

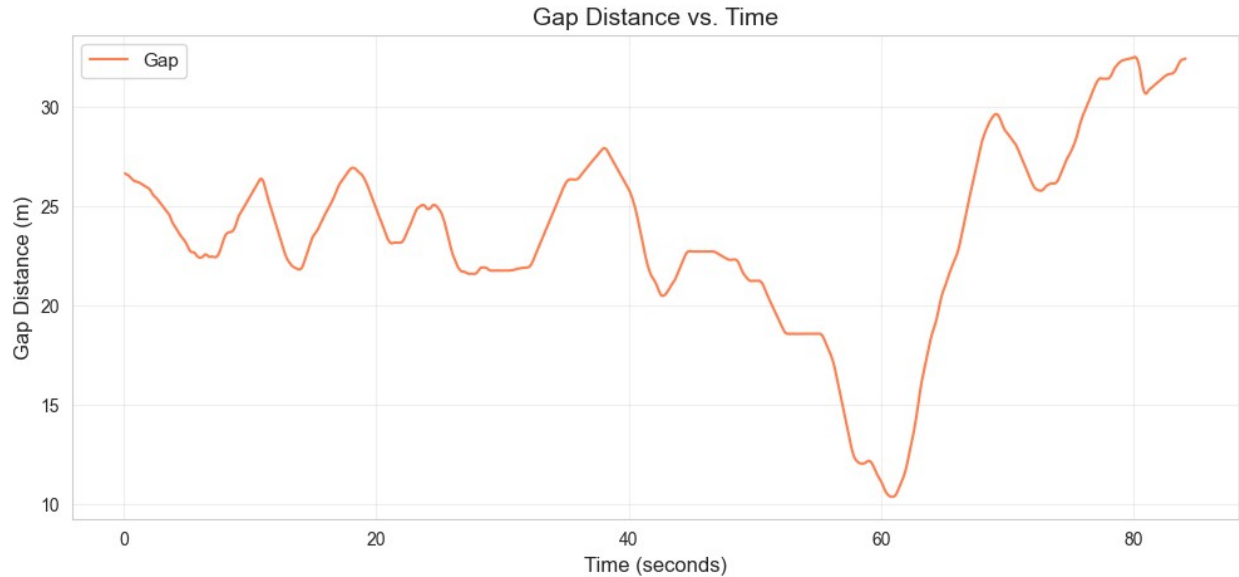


Figure 5. Observed leader-follower gap distance over time.

The observed spacing between the leader and follower averaged 23.60 m, with a median of 23.70 m. The minimum observed gap was 10.36 m and the maximum was 32.53 m. The gap therefore varied by more than 22 m over the trajectory, showing that the follower did not maintain a fixed spacing.

From a traffic-flow perspective, this variation is expected. Drivers continuously adjust spacing in response to changes in leader speed, relative speed, perceived comfort, and reaction behavior. The gap-time plot therefore provides the physical context needed to understand why a dynamic car-following model such as IDM is more appropriate than a fixed-spacing rule.

Trajectory Metric	Value
Observations	841
Time range	0.1–84.1 s
Mean gap	23.60 m
Median gap	23.70 m
Minimum gap	10.36 m
Maximum gap	32.53 m
Mean leader speed	7.44 m/s
Mean follower speed	7.38 m/s

6. Intelligent Driver Model Simulation

6.1 Simulation Setup

The simulation used the observed leader position and speed from trajectory pair 1 as the external forcing input. The simulated follower was initialized at the real follower's initial position and speed, then evolved forward using the IDM acceleration equation and Euler integration. This structure makes the comparison meaningful because the model is asked to reproduce follower behavior while experiencing the same observed leader trajectory.

IDM Parameter	Value	Interpretation
Desired speed, v_0	30 m/s	Free-flow target speed
Minimum spacing, s_0	2 m	Minimum desired gap
Time headway, T	1.5 s	Desired following time gap
Maximum acceleration, a	1.0 m/s ²	Acceleration capability
Comfortable deceleration, b	1.5 m/s ²	Preferred braking intensity
Acceleration exponent, δ	4	Controls approach to desired speed
Time step, Δt	0.1 s	Euler integration interval

6.2 Simulated vs. Observed Follower

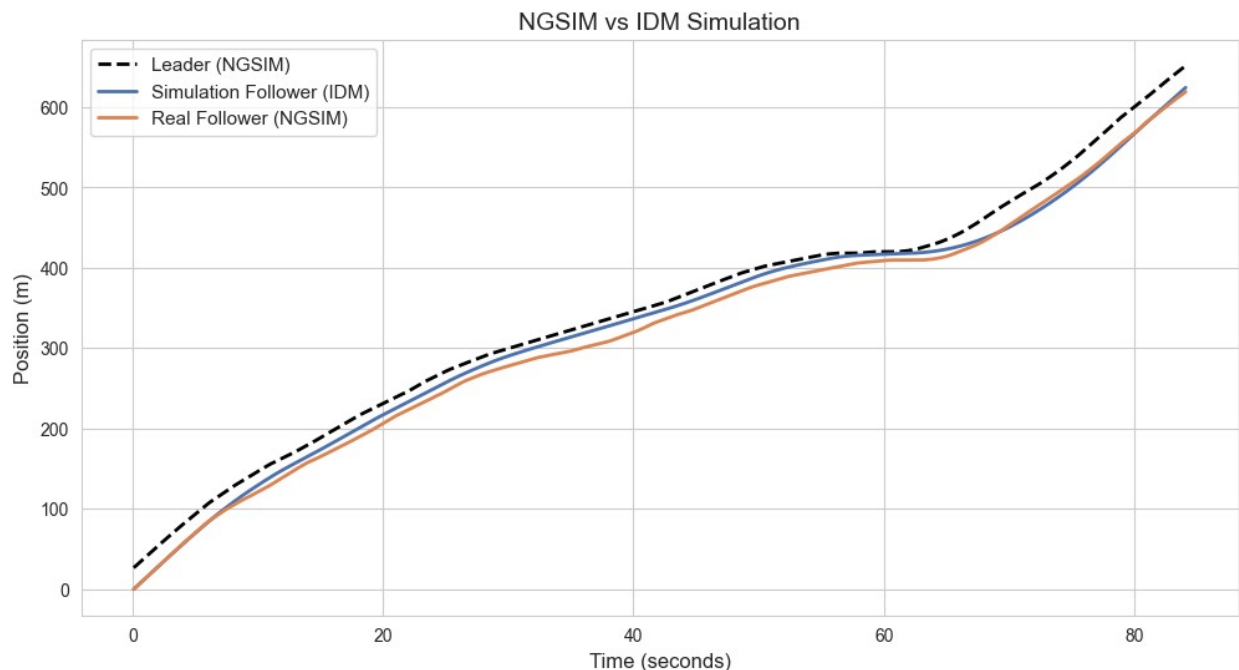


Figure 6. *NGSIM leader and observed follower compared with the IDM-simulated follower.*

The simulated follower follows the same overall movement trend as the real follower and remains behind the leader throughout the plotted trajectory. Visually, the model captures the large-scale stop-and-go pattern but produces a smoother response than the observed human-driven trajectory. This difference is consistent with the deterministic structure of IDM: given the same state and fixed parameters, the model produces a smooth mathematical response, whereas real drivers exhibit reaction variability, imperfect perception, changing preferences, and other unmodeled behavior.

For report-level evaluation, additional error metrics were calculated from the same simulation outputs. Position RMSE was approximately 9.91 m, mean absolute position error was 8.66 m, and the maximum absolute position error was 19.13 m. At the final observation, the simulated follower was 5.30 m ahead of the observed follower. Simulated versus observed follower speed had an RMSE of approximately 1.23 m/s.

The simulated and observed positions had a very high linear correlation of approximately 0.999. However, correlation alone should not be interpreted as evidence of perfect model accuracy because both trajectories share a strong increasing position trend over time. The error metrics reveal meaningful spacing between the simulated and observed trajectories despite the high correlation. This is an important modeling lesson: visual agreement and correlation should be evaluated together with absolute error measures.

Simulation Performance Metric	Value
Position RMSE	9.91 m
Position MAE	8.66 m
Maximum absolute position error	19.13 m
Final position error (simulated – observed)	+5.30 m
Follower speed RMSE	1.23 m/s
Position correlation	0.999
Speed correlation	0.952

6.3 Interpretation of Model Performance

The purpose of the IDM exercise was not to produce a fully calibrated research-grade traffic model; it was to demonstrate how a physically interpretable car-following model can be implemented and compared with real trajectory data. The model succeeded in reproducing the broad direction and timing of vehicle movement, but its remaining error shows that a single fixed parameter set is not sufficient to reproduce every detail of one observed driver's behavior.

A logical next step would be parameter calibration. Values such as desired speed, time headway, acceleration, and comfortable deceleration could be optimized against multiple NGSIM trajectory pairs using an error objective such as RMSE. Calibration would convert the current proof-of-concept simulation into a stronger empirical modeling exercise and would also allow comparison of driver heterogeneity across trajectories.

7. Transportation Engineering Significance

The project connects three levels of transportation analysis. NHTS data provides household and vehicle context, NGSIM provides second-by-second microscopic traffic behavior, and IDM translates observed traffic behavior into a mathematical simulation framework. Together, these components illustrate how transportation engineers move between descriptive data analysis and predictive modeling.

- Travel-demand and planning applications: NHTS-style data can help describe household vehicle availability, fleet characteristics, and differences between urban and rural contexts.
- Traffic operations applications: NGSIM trajectories support analysis of spacing, speed, acceleration, queueing, and car-following behavior.

- Microscopic simulation applications: IDM provides a transparent model for testing how desired speed, headway, acceleration, and braking assumptions influence traffic trajectories.
- Connected and automated vehicle research: car-following models provide a baseline for studying automated longitudinal control and traffic stability.
- Safety and capacity studies: gap and speed behavior are fundamental to understanding rear-end risk, lane-flow characteristics, and roadway performance.

8. Coding Decisions, Bias, and Limitations

8.1 NHTS Visualization Decisions

The NHTS results depend on which variables and plot types were selected. Using raw counts for vehicle type describes the sample directly, but it does not apply survey weights. Similarly, the urban/rural histogram uses counts rather than normalized density, so differences in sample size affect bar heights. These choices are appropriate for an introductory visualization exercise but should be reconsidered before drawing population-level conclusions.

8.2 Selection of a Single NGSIM Trajectory

The final notebook selected trajectory pair 1. This makes the workflow easy to understand and reproduce, but the behavior of one pair cannot represent the full range of drivers in the 16 available trajectories. A different trajectory could have different speeds, gaps, accelerations, and model errors. Any general statement about IDM performance should therefore be based on multiple trajectory pairs.

8.3 IDM Parameter Selection

The IDM parameters were specified directly rather than calibrated from data. The chosen values create a plausible simulation, but they embed assumptions about desired speed, minimum spacing, time headway, acceleration, and comfortable deceleration. Changing these values would change the simulated trajectory. This is a form of modeling bias: the output is partly determined by analyst-selected parameters rather than observed driver-specific calibration.

8.4 Numerical Safeguards and Euler Integration

The code uses a minimum gap of 0.1 m to avoid zero or negative spacing in the IDM denominator and constrains simulated speed to be nonnegative. These are sensible numerical safeguards, but they alter the mathematical model at extreme states. The position update also uses first-order Euler integration. With a 0.1-second time step, this is adequate for demonstration, but more accurate integration schemes could be considered for research-quality simulation.

8.5 Data Quality and Missing Information

The NGSIM file contained no missing values in the supplied data. The NHTS file contained missing values in some columns, including 116 missing vehicle-make entries and 32 missing fuel-type entries. The final notebook's principal visualizations use variables that were otherwise complete. Still, missing categories and unequal group sizes should be documented because they can influence visual interpretation.

8.6 Ethical and Professional Interpretation

Transportation models can influence planning, policy, and operational decisions. Engineers therefore have a responsibility to distinguish descriptive findings from causal claims, document model assumptions, and avoid presenting one trajectory or one parameter set as universally representative. The value of Python is not that it removes engineering judgment; it makes assumptions reproducible and easier to inspect.

9. Reproducibility and Repository Use

The project is well suited for GitHub because the complete workflow can be reproduced from the notebook and two CSV files. A user can place NHTS.csv and NGSIM.csv in the same working directory as the notebook, install pandas, NumPy, Matplotlib, and Seaborn, and run the cells sequentially. The notebook then recreates all six visualizations and the IDM simulation.

For portfolio use, the repository should include the Python notebook, NHTS.csv, NGSIM.csv, the scope of work, this report, and a README that explains the project goal, methods, files, and key results. Including exported figures separately is also helpful because GitHub visitors can view results immediately without executing the notebook.

10. Conclusions and Recommendations

Project #3 successfully combined household-level transportation data analysis with microscopic vehicle simulation. The NHTS analysis showed that automobiles, SUVs, and pickup trucks dominated the supplied vehicle records and that rural vehicles in the sample were, on average, older than urban vehicles. The make-based boxplot further demonstrated that vehicle-age distributions vary widely across manufacturers and contain substantial within-make variability.

The NGSIM analysis showed a dynamic leader-follower relationship in which spacing varied between 10.36 m and 32.53 m over trajectory pair 1. Leader and follower mean speeds were similar, while the time-series plots showed the short-term adjustments that define car-following behavior.

The IDM simulation reproduced the broad follower trajectory but did not perfectly match observed human driving. This is an expected and useful result: a deterministic car-following model captures major physical relationships but simplifies driver heterogeneity and behavioral variation. The project therefore demonstrates both the value and limitations of mathematical traffic models.

Future work should expand the analysis to all available NGSIM trajectories, calibrate IDM parameters empirically, compare alternative car-following models, normalize or weight NHTS analyses appropriately, and add quantitative model-validation plots such as error-versus-time and observed-versus-simulated speed. These improvements would turn the current educational workflow into a stronger transportation research analysis.

References

Federal Highway Administration (FHWA). National Household Travel Survey (NHTS).

Federal Highway Administration (FHWA). Next Generation Simulation (NGSIM) trajectory data.

Treiber, M., Hennecke, A., & Helbing, D. Intelligent Driver Model (IDM) car-following framework.

Python Software Foundation. Python documentation.

pandas Development Team. pandas documentation.

NumPy Developers. NumPy documentation.

Matplotlib Development Team. Matplotlib documentation.

Seaborn Development Team. Seaborn statistical data visualization documentation.

Appendix A – Key Numerical Results

Category	Metric	Result
NHTS	Records	14,684
NHTS	Urban records	11,084
NHTS	Rural records	3,600
NHTS	Top 3 vehicle categories	90.83% of records
NHTS	Urban mean vehicle age	9.67 years
NHTS	Rural mean vehicle age	11.72 years
NGSIM	Total observations	8,166
NGSIM	Trajectory pairs	16
Trajectory 1	Observations	841
Trajectory 1	Mean gap	23.60 m
Trajectory 1	Gap range	10.36–32.53 m
IDM	Position RMSE	9.91 m
IDM	Speed RMSE	1.23 m/s
IDM	Final position error	+5.30 m